Programmable Networking: SDN Foundations

Raouf Boutaba

David R. Cheriton School of Computer Science

University of Waterloo

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SDN in a nutshell

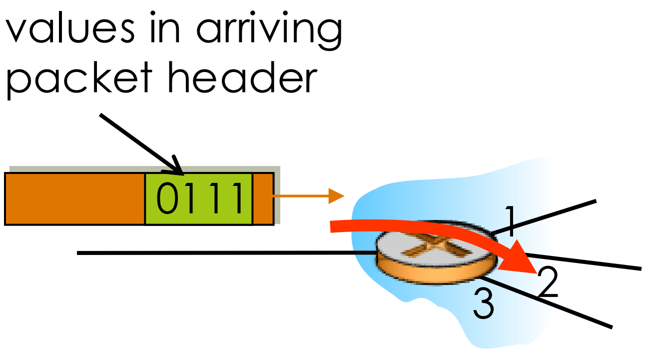
- Three principles:
  - decoupling of control and data planes
  - implementing control plane in software as a (logically) centralized server
  - processing network traffic as flows

Network Planes

- Data Plane
  - responsible for forwarding packets
- Control Plane
  - responsible for defining the behavior of the network
- Management Plane
  - configures routing protocol, monitors network devices and handle anomalies

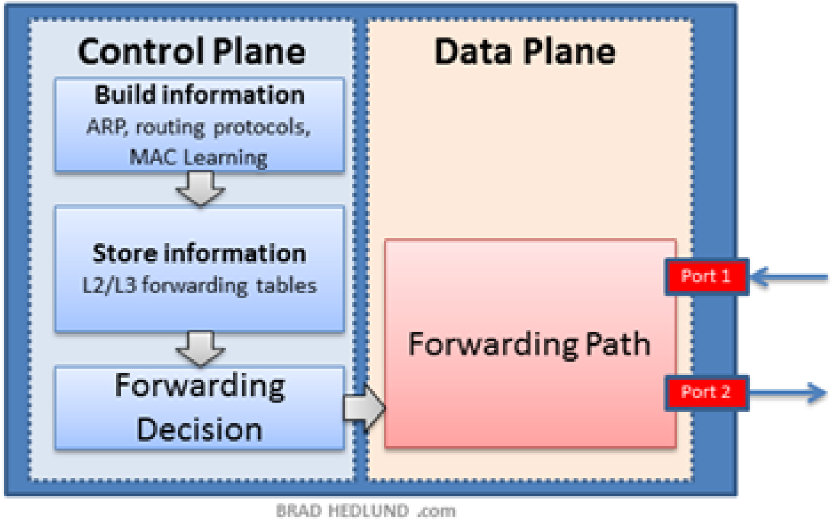
Data plane

- Responsible for moving packets
  - local, per-router function
  - determines how packet arriving on router input port is forwarded to router output port
  - forwarding function
- Other data plane functions:
  - Packet scheduling and buffering
  - Packet fragmentation
  - Bandwidth metering
  - Access control
  - Packet cloning for multicasting
  - Traffic monitoring



Control plane

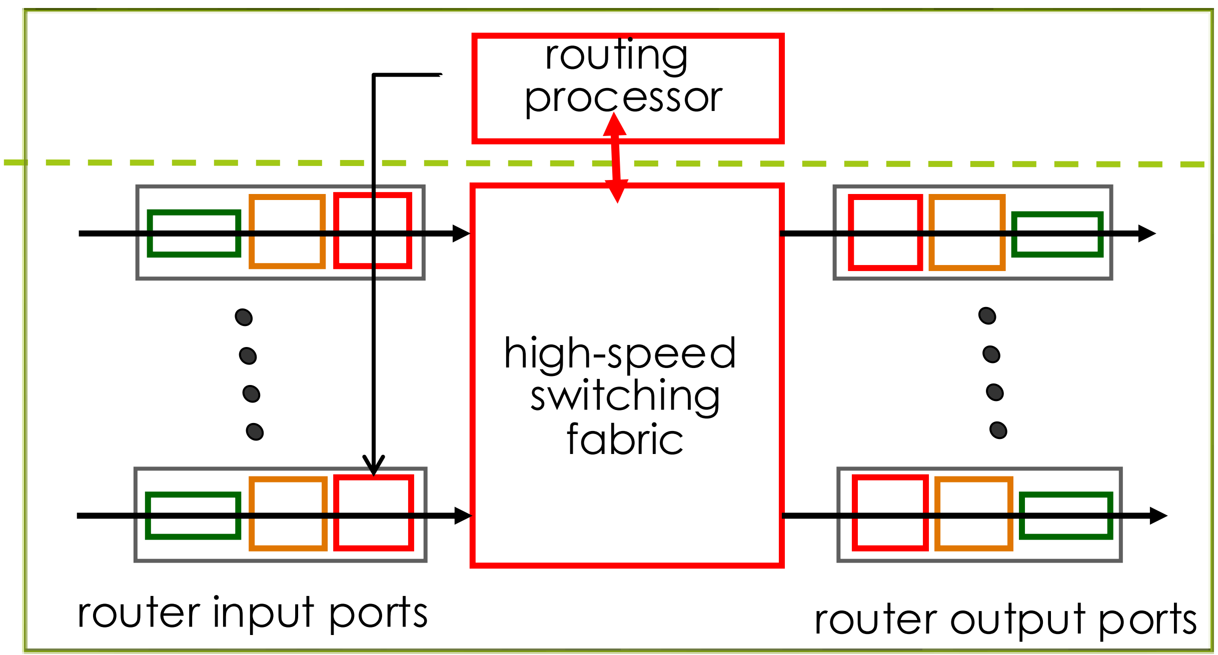
- Defines the network behavior
  - network-wide logic
  - determines how packet is routed among routers along end-end path from source host to destination host
  - Route computation function



- Other control plane functions:
  - FIB installation in routers
  - Topology discovery
  - Access control list generation
  - QoS enforcement

Traditional Networking

- Control plane is vertically integrated with data plane
  - Each router runs its own control plane
- Generic router architecture:

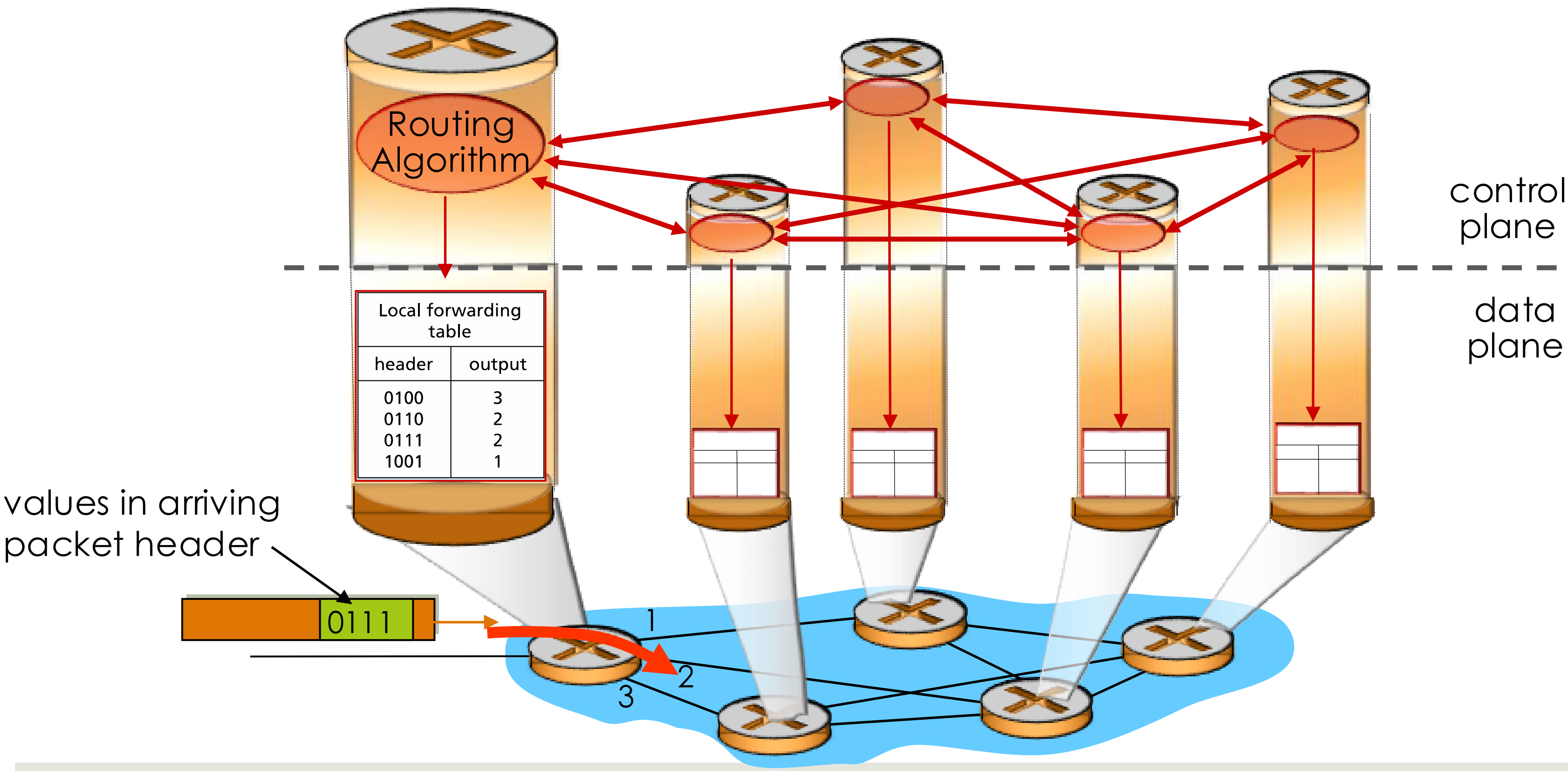


routing control plane (software) operates in millisecond time frame

forwarding data plane (hardware) operates in nanosecond timeframe

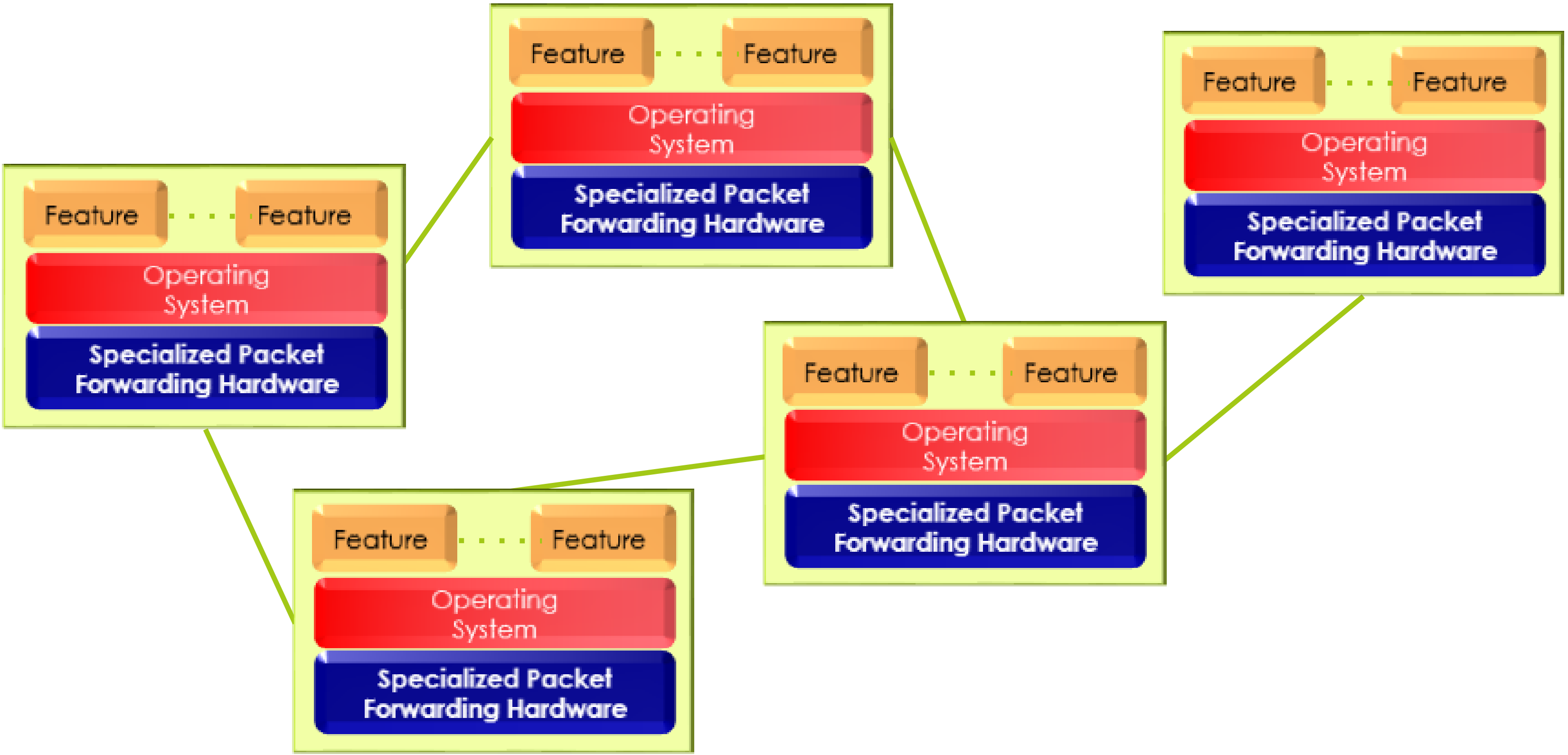
Traditional Routing

- Traditional routing protocols are distributed
  - Routers exchange connectivity info. with each other
  - Each router builds its own routing table
  - Forwarding tables (FIB) are populated based on information available in the routing table



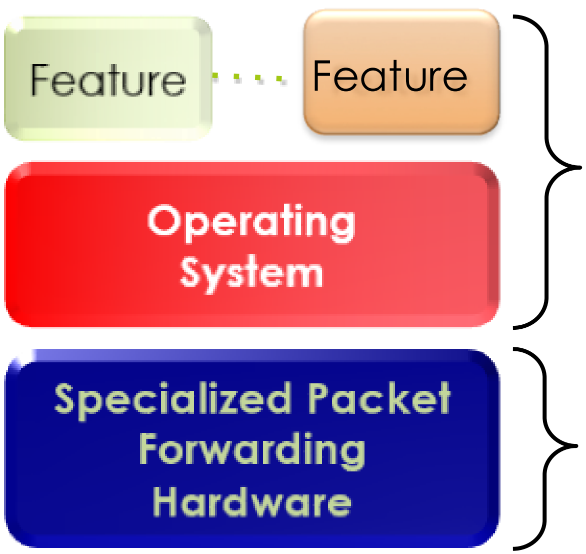
Distributed Control Plane

- Distributed per-router control plane
  - Vertically integrated router contains switching hardware, runs proprietary implementation of Internet standard protocols (IP, RIP, IS-IS, OSPF, BGP) in proprietary router OS (e.g., Cisco IOS)



Traditional Network Element

Routing, management, mobility management, access control, VPNs, …



Millions of lines of code — 8298 RFCs — Barrier to entry

Billions of gates — Complex — Power Hungry

Shortcomings of traditional networking

- Lack of programmability
  - Changes in the network requires manual reconfiguration
  - Error-prone and time consuming
  - Network becomes hard to scale
- Vendor dependency
  - Handful of vendors have monopoly in the market
  - Lack of open interface
  - Software vertically integrated with hardware
- Over specified
  - Slow protocol standardization

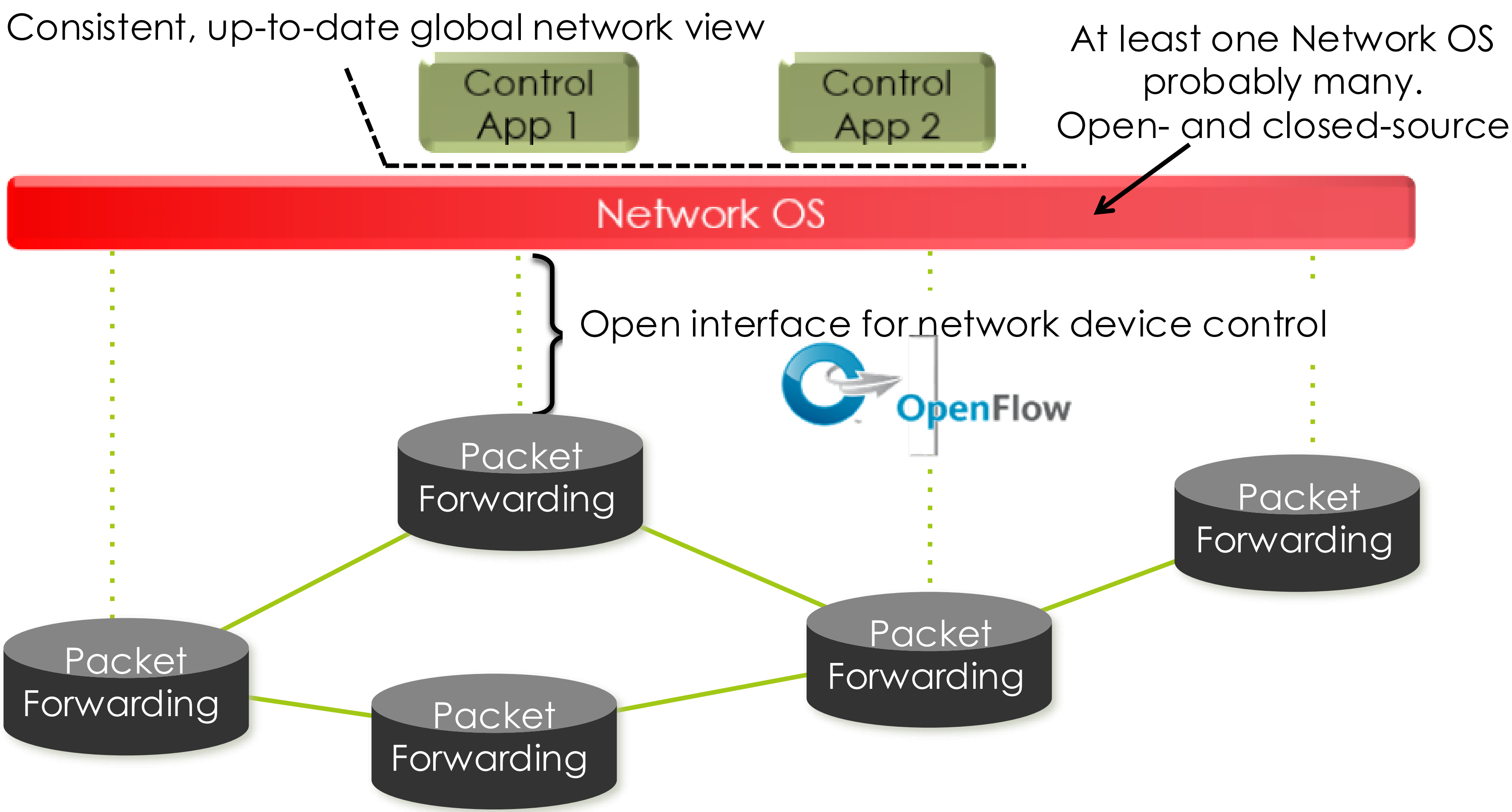
Shortcomings of traditional networking

- Less opportunities for innovation
  - Only vendors can write the code
  - Slow time-to-market for new services/features
- High operational cost
  - More than 50%
  - Human error still causes most outages
- Buggy software
  - Hardware with 20+ million lines of code
  - Cascading failures, vulnerabilities, *etc*.

The advent of SDN

- ~2005: renewed interest in rethinking network control plane
- Software Defined Networking (SDN) emerged as a new networking architecture that
  - Separates control plane from data plane
  - Software running on a *logically centralized* server remotely controls network hardware
  - Open standards (vendor neutral)
  - Directly programmable
  - Facilitates automation & programmability

SDN architecture



Why SDN for future networks?

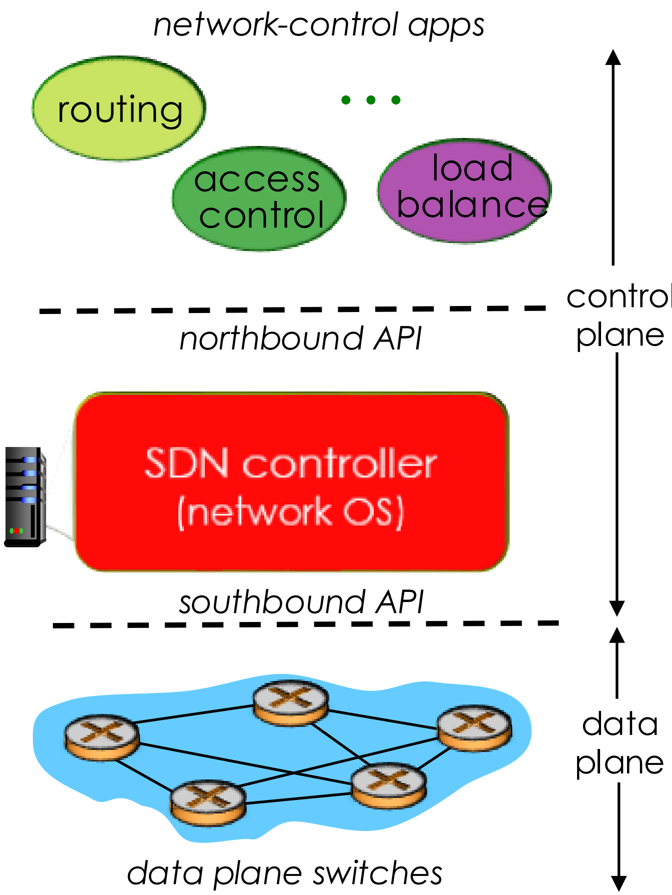
- Separation of control and data planes
  - Enables independent development of new features without hardware modification
- Global network view at control plane
  - Network-wide view allows easier application of policies
- Management Automation
  - Minimize human intervention
  - Lower operational cost
- Monitoring
  - On-demand monitoring of resources
  - Dynamically change what to monitor
  - Application performance can be improved by advanced monitoring features

Why SDN for future networks?

- Elasticity
  - Resources can be scaled up or down on-demand
- More Flexible/Easier Performance Management
  - Traffic engineering
  - Capacity optimization
  - Load balancing
  - Faster failure recovery
- Network Function Integration
  - On-demand provisioning of network functions
    - Firewalls, IDS, IPS, etc.

Components of SDN

- Data plane switches
  - SDN controlled switches
- South Bound API
  - Interface between control and data plane (e.g. OpenFlow)
- SDN Controller
- North Bound API
  - Interface between controller and control apps



From Theory to Practice

- Traditional networks: control and data plane vertically integrated
- SDN separates them: control plane becomes software
  - Network behaviour defined programmatically
- In Lab 1 you take direct control of the data plane
  - Mininet: network emulator, full topology on one machine
  - OVS: software switch, programmatically controlled
  - Flow rules: tool for defining network behaviour
  - No controller yet: you are the control plane